

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**CSPIT/DEPSTAR****First Semester of B. Tech (EC,CE,IT,DCE,DCSE,DIT) Examination****December 2019****EE 145 Basics of Electronics and Electrical Engineering****Date: 18/12/2019, Wednesday****Time: 10:00 a.m. to 1:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below. [07]**

- i. A dielectric $K = 2$ is inserted between the plates of a $20\mu\text{F}$ capacitor. Its capacitance will become? [01]
(a) $40\mu\text{F}$ (b) $28\mu\text{F}$ (c) $12\mu\text{F}$ (d) $18\mu\text{F}$
- ii. In case the charges are similar the force is of? [01]
(a) Attraction (b) Repulsion (c) Maximum Value (d) Zero Value
- iii. The presence of a dielectric between the plates of a capacitor results in? [01]
(a) increasing its capacitance (b) constant capacitance (c) decreasing its capacitance
(d) Zero capacitance
- iv. The property of a coil by which a counter emf is induced in it when the current through the coil changes is known as [01]
(a) Self Inductance (b) Mutual Inductance (c) Series Aiding Inductance (d) Capacitance
- v. The direction of induced emf can be found by [01]
(a) Laplace's Law (b) Lenz's Law (c) Fleming's left hand rule (d) Kirchhoff's Voltage Law
- vi. In a PN junction when the applied voltage overcomes the..... potential, the diode current is large, which is known as..... [02]
(a) Depletion, negative bias (b) Reverse, reverse bias (c) Resistance, reverse bias
(d) Barrier, forward bias

Q – 2. Answer any two questions. [14]

- A. (i) A parallel plate capacitor has plates of area 1.5m^2 . It has three dielectrics 1mm, 1.6mm and 2mm thick. The relative permittivities of these dielectrics are 2, 4, and 5 respectively. Calculate the capacitance of the capacitor and the electric field strengths in the dielectrics if a voltage of 2.5KV is applied between the plates. [05]
- (ii) Find the approximate equivalent capacitance of the combination of capacitors in the figure 1, where $C_1=5.0\text{ F}$, $C_2=5.0\text{ F}$, $C_3=5.0\text{ F}$, $C_4=10.0\text{ F}$, $C_5=10.0\text{ F}$ and $C_6=10.0\text{ F}$. [2]

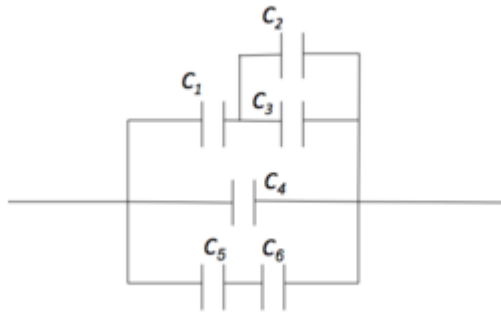


Figure 1

- B. Find node voltages and calculate the current flowing through 20ohm resistor for the [07]
circuit shown in figure 2?

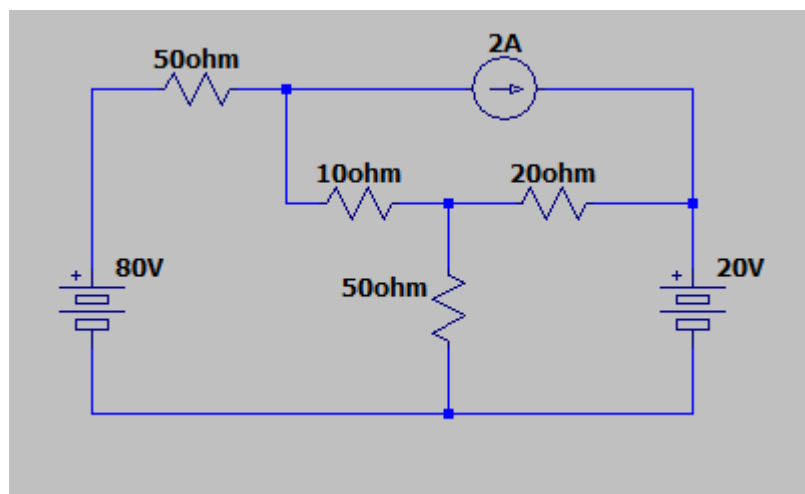


Figure 2

- C. Determine the current through the 10-k resistor in the circuit of Fig. 3 using mesh [07]
analysis.

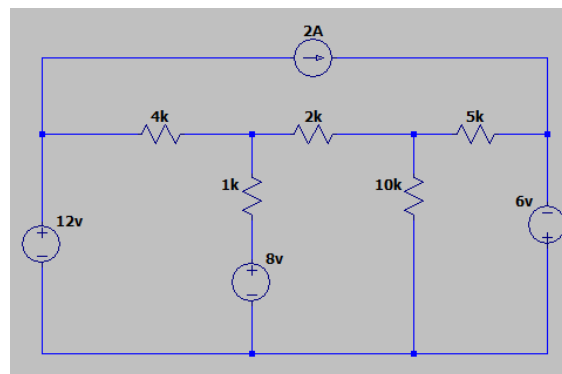


Figure 3

Q – 3. Answer any two questions. [14]

- A. For the common base circuit shown in Fig. 4, determine I_C and V_{CB} . Assume the [07]
transistor to be of silicon.

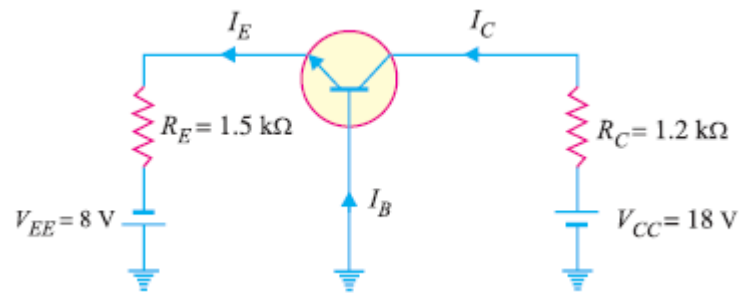


Figure 4

- B.** Explain the working of Common base transistor in detail with necessary equations, circuit diagram and characteristics. [07]
- C.** (I) The circuit of Figure 5 uses two zener diodes, each rated at 15 V, 200 mA. If the circuit is connected to a 45-volt unregulated supply, determine : [04]
- (i) The regulated output voltage (ii) The value of series resistance R

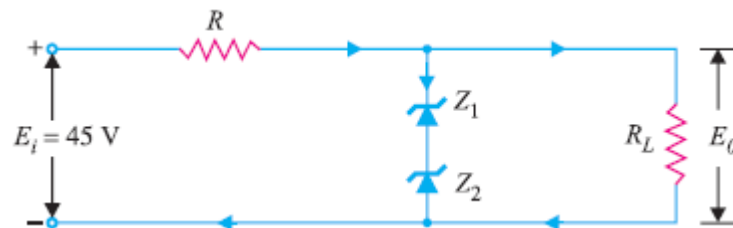


Figure 5

- (II) A 10-V zener diode is used to regulate the voltage across a variable load resistor as shown in figure 6. The input voltage varies between 13 V and 16 V and the load current varies between 10 mA and 85 mA. The minimum zener current is 15 mA. Calculate the value of series resistance R. [03]

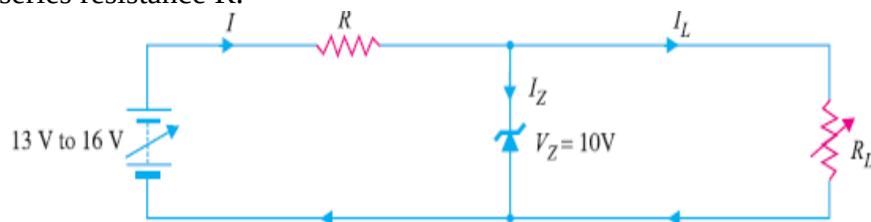


Figure 6

SECTION – II**Q - 4 Answer the questions below. [07]**

- i The SI unit of specific resistance(Resistivity) is [01]
 a. mho-m
 b. ohm-m
 c. mho/m
 d. ohm/m
- ii. A length of wire has a resistance of 6 ohms. The resistance of a wire of the same material three times as long and twice the cross-sectional area will be [01]
 a. 9 ohms
 b. 36 ohms
 c. 1 ohms
 d. 12 ohms
- iii. The basic requirement for inducing e.m.f in a coil is that [01]
 a. Magnetic flux should link the coil
 b. there should be change in magnetic flux linking the coil
 c. coil should form a closed loop
 d. none of the above
- iv A current of 2 A through a coil sets up the magnetic flux linkages of 4 Wb-turn. The inductance of the coils is [01]
 a. 8 H
 b. 0.5 H
 c. 2 H
 d. 1H
- v An a.c. generator having 10 poles and running at 600 r.p.m will generate an alternating voltage of frequency [01]
 a. 25 Hz
 b. 100 Hz
 c. 50 Hz
 d. 200 Hz
- vi The a.c. system is preferred to dc system because [01]
 a. a.c. voltage cannot be easily changed in magnitude
 b. d.c. motors do not have fine speed control
 c. high voltage a.c transmission is efficient
 d. d.c. voltage cannot be used for domestic appliances
- vii The power factor of an a.c circuit is given by [01]
 a. X_L/R
 b. Z/R
 c. R/X_L
 d. R/Z

Q-5 Answer any two questions [14]

- A (i.) The field winding of a generator has a resistance of 12.7Ω at 18°C and 14.3Ω at 50°C . find (a) temperature coefficient at 0°C , (b) resistance at 0°C , (c) temperature coefficient at 18°C [05]

- (ii.) Define a. Self induced e.m.f b. Dynamically induced emf [02]
- B** (i) An air cored solenoid 70 cm in length and of 2100 turns has a radius of 4.5 cm. A second coil of 750 turns is wound upon in the middle part of the solenoid. Find 1. Self-inductance of the coil 2. Mutual inductance between the coils. [04]
(ii) Derive the equation of dynamically induced e.m.f with necessary diagrams [03]
- C** (i) List the advantages of Three Phase system over single phase. [04]
(ii) Define the process of modulation and demodulation [03]
- Q-6 Answer any two questions [14]**
- A** Find the average and root mean square value for the sinusoidal waveform. [07]
- B** Explain series RLC circuit with necessary derivation, waveform and diagrams. [07]
- C** Explain the Apparent power, True power and Reactive power. [07]

ALL THE BEST